TECHNICAL REPORT ON

STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME

(SIWES)

HELD AT

FIRST BANK OF NIGERIA PLC

HEAD OFFICE, MARINA, LAGOS.

BY

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CHAPTER ONE

1.0 INTRODUCTION

1.1 HISTORY OF SIWES

The Students' Industrial Work Experience Scheme (SIWES) is a skills training program designed to expose and prepare students of universities, polytechnics, and colleges of education for the industrial work situation they are likely to meet after graduation. The scheme was established by the Federal Government of Nigeria in 1973 under the Industrial Training Fund (ITF) Decree No. 47 of 1971.

The need for the establishment of the scheme arose from the growing concern among industrialists that graduates of Nigeria's higher institutions lacked adequate practical background studies preparatory for employment in the industries. Therefore, employers were of the opinion that the theoretical education provided in higher institutions was not responsive to the needs of the employers of labor. This creates a gap between theory and practice, which SIWES seeks to bridge.

Initially, the ITF funded the scheme entirely. However, as the number of participating institutions and students increased, the financial burden became too heavy for the ITF alone. Consequently, the Federal Government, through the National Universities Commission (NUC) and the National Board for Technical Education (NBTE), stepped in to co-fund the scheme. Today, SIWES is a generally accepted requirement for graduation in most science, technology, and engineering disciplines in Nigerian tertiary institutions.

1.2 AIMS AND OBJECTIVES OF SIWES

The specific objectives of SIWES are to:

- Provide an avenue for students in Nigerian universities to acquire industrial skills and experience in their course of study.
- Prepare students for the work situation they are likely to meet after graduation.
- Expose students to work methods and techniques in handling equipment and machinery that may not be available in the universities.
- Make the transition from the university to the world of work easier and thus enhance students' contacts for later job placement.

- Provide students with an opportunity to apply their theoretical knowledge in real work situations, thereby bridging the gap between university work and actual practice.
- Enlist and strengthen employers' involvement in the entire educational process of preparing university graduates for employment in industry.

1.3 BODIES INVOLVED IN THE MANAGEMENT OF SIWES

The bodies involved in the management of the Students' Industrial Work Experience Scheme in Nigeria are:

1. **The Federal Government:** Provides funds/grants for the scheme through the Ministry of Commerce and Industry.
2. **The Industrial Training Fund (ITF):** Formulates policies and guidelines on SIWES distribution of funds to institutions.
3. **Supervising Agencies (NUC, NBTE, NCCE):** Ensure that institutions establish functioning SIWES units and appoint coordinators.
4. **Tertiary Institutions:** Prepare students for industrial attachment and place them in relevant industries.
5. **The Employers of Labor:** Accept students for industrial training and provide necessary facilities for their training.

CHAPTER TWO

2.0 THE ORGANIZATION: FIRST BANK OF NIGERIA PLC

2.1 HISTORICAL BACKGROUND

First Bank of Nigeria Limited (FirstBank) is the premier bank in West Africa with its impact woven into the fabric of society. The Bank has been sustaining its development-oriented services for over 130 years as the region's foremost financial inclusion services provider. FirstBank was established in 1894 as the Bank of British West Africa (BBWA) by Sir Alfred Jones, a shipping magnate from Liverpool. It commenced operations with a humble capitalization, operating from a small office in Marina, Lagos, which remains its headquarters today.

Throughout its history, the bank has undergone several name changes reflecting the political and economic evolution of Nigeria. In 1957, it became the Bank of West Africa (BWA). Following Nigeria's independence and subsequent localization policies, it was incorporated locally as the Standard Bank of Nigeria Limited in 1969. In 1979, the name was changed to First Bank of Nigeria Limited, and later to First Bank of Nigeria Plc in 1991.

Today, FirstBank boasts over 800 business locations and over 200,000 Banking Agents spread across 99% of the 774 Local Government Areas in Nigeria. It specializes in Retail Banking, Corporate Banking, Commercial Banking, and Public Sector Banking, maintaining its leadership position in the financial sector.

2.2 ORGANIZATIONAL STRUCTURE

The bank is structured to ensure efficiency and effective service delivery. The hierarchy ensures that decisions flow from the Board of Directors down to the operational staff. Below is a simplified representation of the organizational structure relevant to my placement.

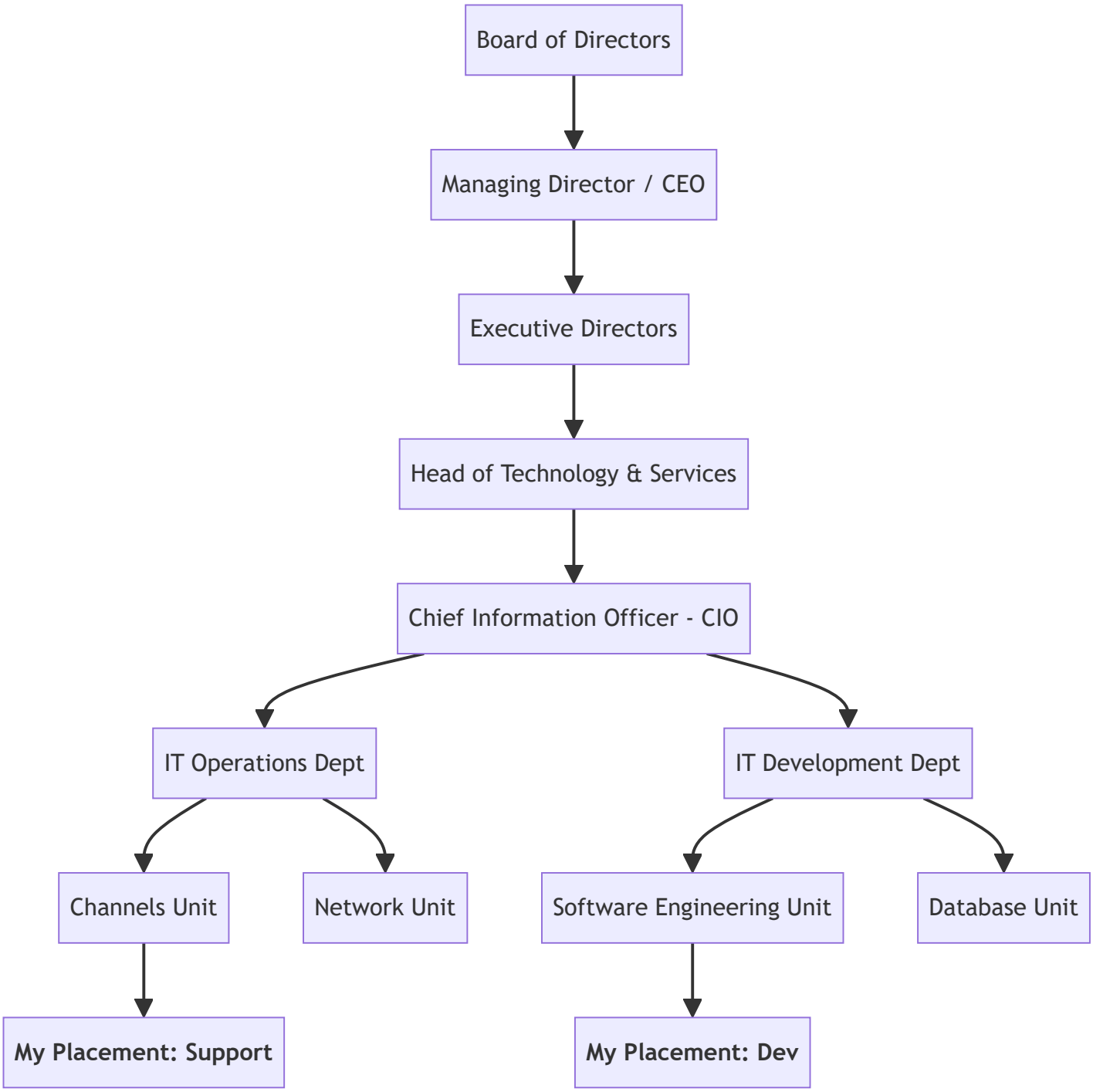

Figure 2.1: Simplified Organogram of First Bank IT Department

2.3 THE INFORMATION TECHNOLOGY DEPARTMENT

The Information Technology (IT) Department is the backbone of First Bank's operations, given that modern banking is entirely dependent on digital platforms. The department is tasked with maintaining the bank's core banking application (Finacle), managing ATMs, supporting mobile and web banking channels, and developing proprietary software solutions.

During my SIWES program, I had the privilege of rotating through two key units within this department:

2.3.1 The Channels Unit

This unit is responsible for ensuring the availability and reliability of the bank's electronic banking channels, including ATMs, POS terminals, and mobile apps. They handle second-level support issues, investigating transaction failures that front-line customer service cannot resolve. Speed and accuracy are paramount here, as downtime translates directly to financial loss and reputational damage.

2.3.2 The Development Unit

This unit functions as an internal software house. They build and maintain custom applications that bridge gaps between the core banking system and external services. They utilize modern enterprise technologies such as Java Enterprise Edition (Java EE) and Spring Boot for backend services, and various frontend frameworks for user interfaces. This environment is fast-paced and requires disciplined coding practices.

CHAPTER THREE

3.0 WORK EXPERIENCE ACQUIRED

This chapter gives a detailed account of the technical activities I undertook during my six-month industrial training. My experience was divided into two major phases: Support Operations in the Channels Unit and Software Engineering in the Development Unit.

3.1 PHASE 1: CHANNELS UNIT (SUPPORT & DATABASE MANAGEMENT)

For the first three months of my placement, I worked in the Channels Support Unit under the supervision of Mr. Babatunde. This unit serves as the "engine room" for resolving complex transaction issues that occur on the bank's digital platforms (FirstMobile, USSD *894#, and Internet Banking).

3.1.1 Daily Routine and Workflow

My daily tasks involved monitoring the ticket queue on the bank's CRM software. Issues ranged from "debit without credit" scenarios to "failed bill payments." My role was to investigate the true status of these transactions by querying the database directly.

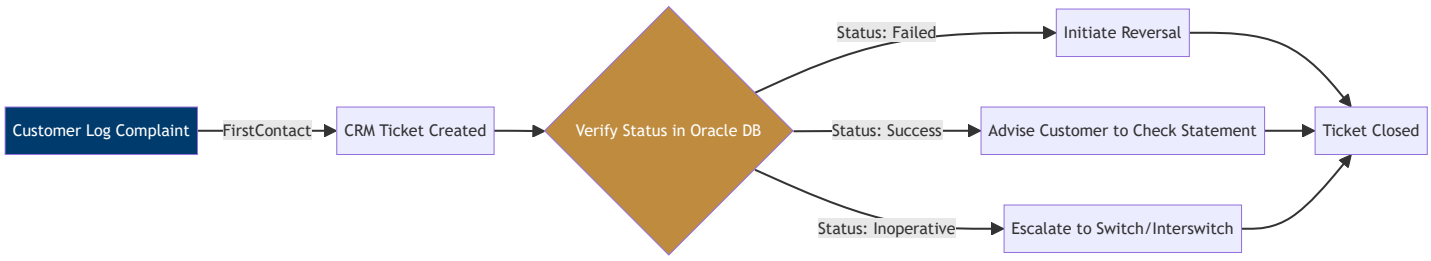

Figure 3.1: Channels Unit Issue Resolution Workflow

The workflow was as follows:

1. Receive Ticket from Customer Care (FirstContact).
2. Extract the unique `Transaction_Reference_ID`.
3. Connect to the Oracle Database using **Toad for Oracle**.
4. Run SQL `SELECT` queries to check the status in the `ISW_TRANSACTIONS` table.

5. If status is '00' (Success) but value was not given, initiate a credit reversal.
6. If status is '91' (Issuer Inoperative), escalate to the switch team (Interswitch/NIBSS).

3.1.2 Technical Skill: Structured Query Language (SQL)

The most significant skill I acquired here was proficiency in SQL. I moved from writing basic queries to constructing complex joins to correlate data across multiple tables (e.g., linking the `Account_Master` table with the `Transaction_Log` table).

Below is a representation of a standard investigative query I used daily.

```sql
-- Plate 3.1: SQL Query for Transaction Investigation

SELECT
    t.tran_date, t.reference_id, c.account_number,
    c.account_name, t.amount, t.response_code
FROM
    tbl_transactions t
JOIN
    tbl_customer_accounts c ON t.acct_id = c.acct_id
WHERE
    t.tran_date BETWEEN TO_DATE('2025-01-01', 'YYYY-MM-DD')
    AND TO_DATE('2025-01-02', 'YYYY-MM-DD')
    AND t.response_code != '00';
```

Plate 3.1: Investigative SQL Query used in Channels Unit

3.2 PHASE 2: DEVELOPMENT UNIT (SOFTWARE ENGINEERING)

Upon satisfactory completion of my rotation in the Channels Unit, I was transferred to the Development Unit. This was the highlight of my training, as I was exposed to the full Software Development Life Cycle (SDLC). I worked on two major projects: **Prime-Stack Bank (PSB)** and **BetaAssist**.

3.2.1 Project A: Prime-Stack Bank (PSB) Architecture

PSB is a Customer Information Management System. We utilized the Model-View-Controller (MVC) architecture to ensure separation of concerns.

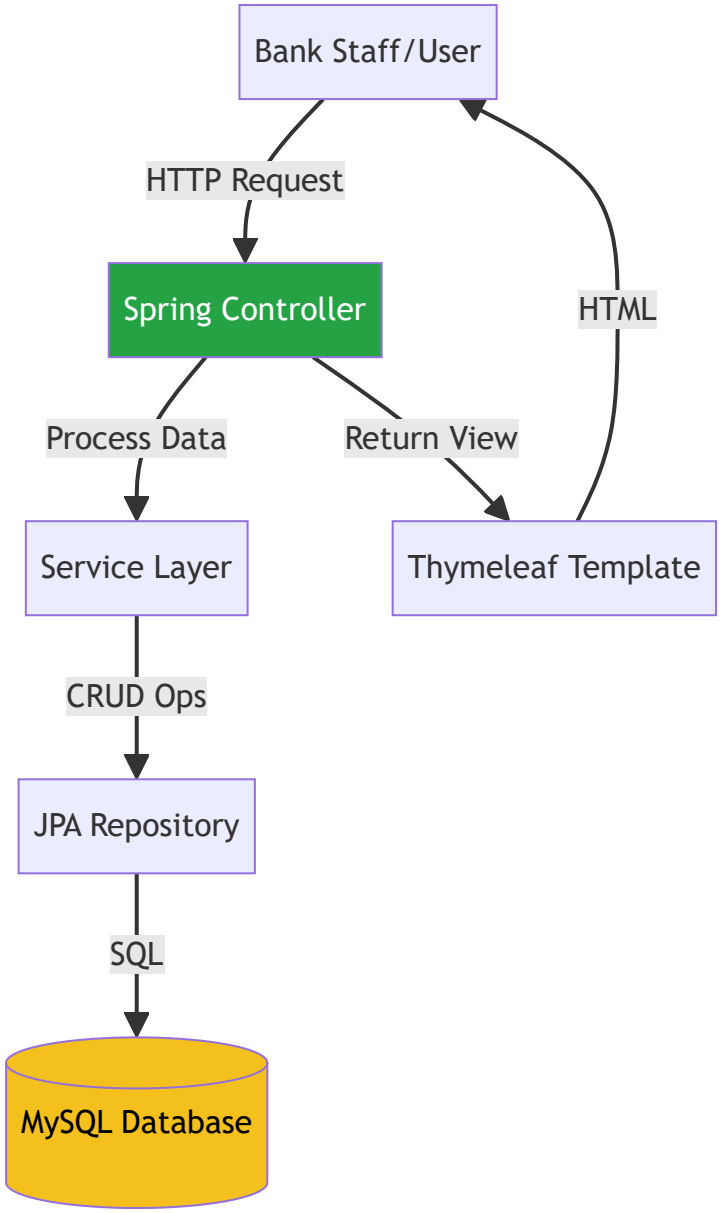

Figure 3.2: PSB System Architecture (MVC Pattern)

Code Snapshot: Controller logic for Customer Onboarding.

```
// Plate 3.2: Java Controller for Customer Registration

@Controller
@RequestMapping("/psb")
public class CustomerInfo {
    @Autowired
    CustomerRepo repo;

    @PostMapping("/save_customer_info")
    public String saveCustomer(@ModelAttribute Customer customer) {
        if(repo.existsById(customer.customerId)){
            return "redirect:/psb/error";
        } else {
```

```
            repo.save(customer);
            return "redirect:/psb/success";
        }
    }
}
```

Plate 3.2: Spring MVC Controller handling Customer Data persistence

3.2.2 Project B: BetaAssist Architecture (On-Device AI)

The Vision: BetaAssist was conceived to bridge the digital divide for elderly and visually impaired customers who find modern mobile banking interfaces complex. The goal was to create an "invisible interface" where a user could simply say, "Transfer 5000 naira to my son," and the agent would execute the task on their behalf.

Technical Implementation - Gemma 3 (On-Device):
Unlike cloud-based assistants (like Siri or Alexa) that require internet, BetaAssist operates entirely offline to ensure data privacy and functionality in low-connectivity areas. We selected the **Gemma 3 (2B parameter)** model for its balance of performance and efficiency.

To run this LLM on a standard Android phone, I had to perform **Model Quantization**. I converted the standard 32-bit floating-point weights into 8-bit integers (int8). This reduced the model size from ~4GB to just 1.5GB, allowing it to load into the RAM of mid-range devices without crashing the operating system.

The Accessibility Service Bridge:
The "eyes" and "hands" of the system relies on the Android Accessibility API. As shown in the code snippet below, the app requests permission to read the screen content of other apps (like the FirstMobile app). When the LLM determines that a button needs to be clicked, the Accessibility Service programmatically dispatches a gesture event to the specific X,Y coordinates of that UI element.

```
// Plate 3.3: Kotlin Logic for AI Command Execution

fun processCommand(json: String){
    val command = Json.decodeFromString<Command>(json)
    when(ActionType.valueOf(command.function)){
        ActionType.openApp -> openApp(command.appName)
        ActionType.extractText -> extractText()
        ActionType.conversation -> conversation(command.prompt!!)
    }
}
```

Plate 3.3: Kotlin Function processing AI-generated JSON commands

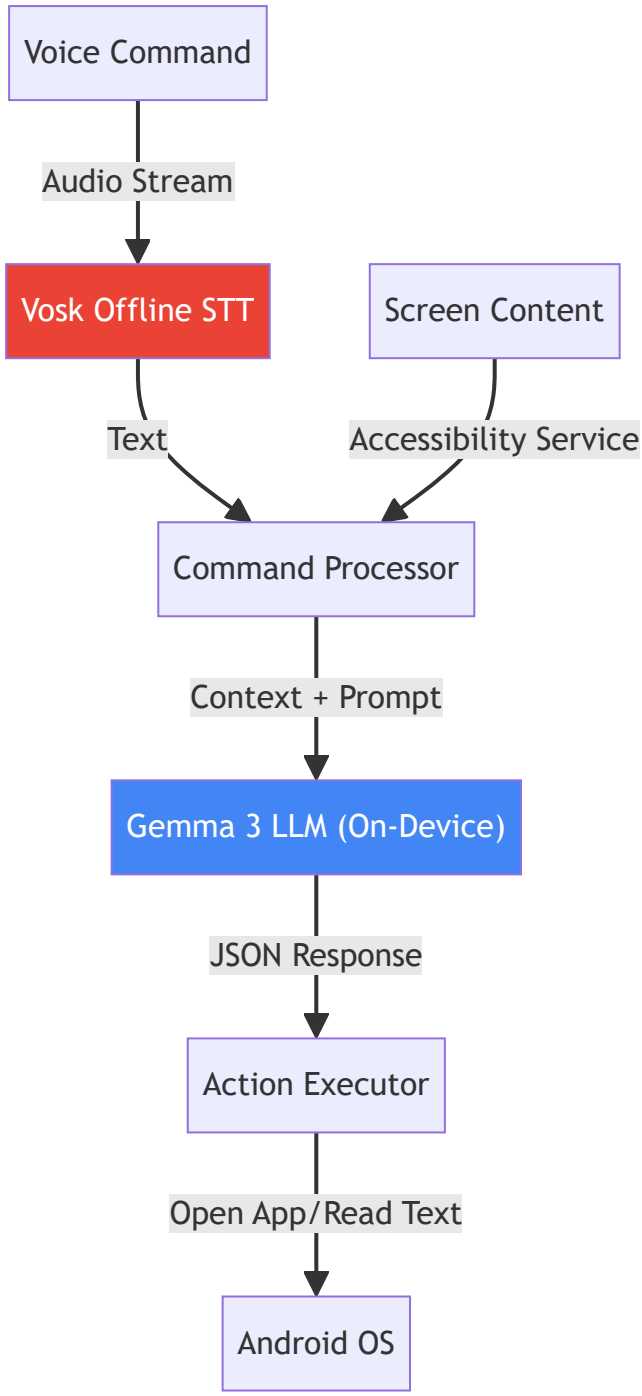

Figure 3.3: BetaAssist Offline AI Architecture

CHAPTER FOUR

4.0 CHALLENGES ENCOUNTERED AND SOLUTIONS

No industrial training is devoid of challenges. These hurdles are, in fact, part of the learning process. During my stay at First Bank, I encountered both technical and operational challenges.

4.1 TECHNICAL CHALLENGES

1. Legacy Code Integration: In the Development Unit, I had to work with some older Java codebases that lacked documentation. Understanding the logic without comments was difficult.
Solution: I learned to use debugging tools in IntelliJ IDEA to step through the code line-by-line to understand the execution flow. I also consulted senior developers for historical context.

2. Android Permission Handling: While building BetaAssist, I faced continuous crashes due to Android's strict permission system for Accessibility Services and Microphone access.
Solution: I researched the Android Developer Documentation and implemented a robust permission-checking logic that prompts the user gracefully before the app crashes.

4.2 OPERATIONAL CHALLENGES

1. Adaptation to Corporate Culture: Transitioning from a casual student lifestyle to a strict 8:00 AM – 5:00 PM banking environment was initially exhausting.
Solution: I created a rigid sleep schedule and prioritized tasks to ensure I could maintain high energy levels throughout the workday.

CHAPTER FIVE

5.0 CONCLUSION AND RECOMMENDATIONS

5.1 CONCLUSION

The six months I spent at First Bank of Nigeria Plc have been the most professionally defining period of my undergraduate studies. I have successfully bridged the gap between the theoretical computer science taught in class and the practical realities of the industry.

I have mastered the use of **SQL** for data manipulation, **Java Spring Boot** for enterprise backend development, and **Android/Kotlin** for mobile solutions. Moreover, the experience of building "BetaAssist" has given me a niche interest in On-Device AI, which I intend to pursue for my final year project.

5.2 RECOMMENDATIONS

5.2.1 To the University (UniAbuja)

The Department should introduce more practical sessions on modern frameworks like Spring Boot and React, as the industry has moved beyond basic Java and HTML. This will reduce the learning curve for students during SIWES.

5.2.2 To the Industry (First Bank)

I recommend that First Bank creates a structured pathway for high-performing SIWES students to return as Graduate Trainees upon the completion of their university education. Retaining students who have already adapted to the bank's unique culture and technical environment would save significant retraining resources.